

AUTOMF: AUTOMATED ENSEMBLES FOR MULTIFIDELITY FIELD PREDICTION

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ABSTRACT

Learning to predict physical fields requires fine simulation data that can be expensive to obtain. Multifidelity learning combines inexpensive coarse solutions with fine data to improve generalization to new parameter inputs, but the most effective way to use these fidelities varies across problems. We present AUTOMF, an automated ensemble workflow for multifidelity field prediction. We construct its library by adapting neural operators, convolutional networks, and reduced basis models into nine parameter to field surrogates. For each dataset, five additional input and fine solution pairs, reserved from surrogate training, guide model selection and ensemble weighting. The workflow either selects an individual model, weights models according to their individual errors, or jointly fits weights to minimize the combined prediction error. Each model receives one weight that remains fixed across the field and across new inputs, allowing prediction without an additional coarse simulation. We assemble a benchmark of 22 datasets spanning seven classes of partial differential equations (PDEs) and climate emulation, combining our simulations with imported data to compare individual models, additional baselines, and ensembles. Different models lead on different datasets, motivating their combination. Compared with selecting the lowest error individual model, the fitted ensemble improves accuracy on 16 datasets and reduces geometric mean relative L^2 error by 11.6% when averaged across PDE classes and 7.5% when averaged across all datasets.



Figure 1: **An ERA5 example from training data to field prediction.** The input x comprises 12 emissions variables and the target y is a fine ERA5 temperature field (Hersbach et al., 2020, Niu et al., 2024). Five reserved fine examples determine inverse error weights w_m for nine model fields $f_m(x)$, whose weighted average is $\hat{y}(x)$. Each weight applies across the whole field. The training pair shares one input, while predictions use a separate reserved input.

1 INTRODUCTION

Scientific design requires solution fields for many choices of physical parameters. Changing a boundary condition or diffusivity changes the temperature throughout a domain. A surrogate model

learns this map without repeating fine simulations. Multifidelity learning adds cheaper coarse solutions (Kennedy & O’Hagan, 2000, Howard et al., 2023, Niu et al., 2024), leaving a practical choice: which model should use them? Transfer learning, joint training, and coarse field correction use different fidelity relationships, and their accuracy can change as those relationships weaken (Faza et al., 2026).

We train a surrogate library and use additional known fine solutions to fit ensemble weights. AUTOMF compares three rules: select the most accurate individual model, average models according to their individual errors, or fit averaging weights to minimize the combined error. New parameters produce a fine field without another coarse solve (Figure 1). Each dataset has its own fitted ensemble, with one weight per model fixed across inputs and grid points. Automation selects and combines models within prescribed training recipes.

An ensemble increases surrogate computation while reusing expensive field data. For climate modeling, the Regional Aerosol Model Intercomparison Project (RAMIP) reports indicative costs of approximately 100,000 CPU core hours for CESM2.1 and 400,000 for UKESM1 per coupled simulation spanning roughly 36 simulated years (Wilcox et al., 2023, Appendix A). Core hours measure total processor usage. These figures describe RAMIP simulations, not the production cost of our ERA5 archive. Niu et al. (2024, Section 6) similarly describe their extra inference overhead as “small compared with the actual simulators”. When generating fields dominates the budget, additional surrogate computation can therefore be reasonable. AUTOMF reuses available training fields across candidates and the same reserved fine examples across selection and combination rules.

This approach builds on engineering surrogate ensembles (Goel et al., 2007, Acar & Rais-Rohani, 2009, Viana et al., 2009), automated model libraries (Feurer et al., 2015, Erickson et al., 2020), and comparative studies of multifidelity field surrogates (Brunel et al., 2025). Our contribution has three connected parts. First, we construct a library of nine field surrogates by adapting established architectures to transfer, pooled supervision, coarse field correction, and fine only prediction. A common parameter to field interface makes these different uses of training data available for selection and combination. Second, we assemble a benchmark of 22 datasets from our simulations and imported data, documenting their fidelity definitions and comparing models on common evaluation cases within each dataset. Third, AUTOMF connects this library and benchmark through an automated ensemble workflow with an explicitly counted budget of additional fine solutions. The underlying architectural components and ensemble weighting principles are established.

We first compare individual library models with additional baselines, then ask whether ensemble fitting improves on choosing one model. Error analysis, fitting loss controls, and ERA5 climate emulation (Hersbach et al., 2020, Niu et al., 2024) examine when combining predictions helps. This separates library accuracy from reliable selection. Historical results informed development, so these comparisons remain retrospective.

2 BACKGROUND: FROM SURROGATE SELECTION TO FIELD ENSEMBLES

Engineering studies already show why choosing one surrogate can be difficult and why combining several can help. Goel et al. (2007) weight surrogates using error estimates, Acar & Rais-Rohani (2009) and Strömberg (2021) optimize metamodel weights, and Viana et al. (2009) compare cross validation selection, weighted prediction, and library subsets. For functional outputs, Brunel et al. (2025) compare more than a dozen multifidelity surrogates in a common framework. Their findings connect model performance to fidelity correlation, training size, and residual complexity. These studies establish both the ensemble principle and the need for comparisons across problems. We extend this empirical line of work to a surrogate library for fields and measure combination gains under small, explicitly counted fitting budgets.

The weighting rules follow stacked generalization (Wolpert, 1992), stacked regression (Breiman, 1996), and Super Learner (van der Laan et al., 2007). Greedy ensemble selection (Caruana et al., 2004), auto sklearn (Feurer et al., 2015, 2022), and AutoGluon (Erickson et al., 2020) automate selection from broader libraries. Related multifidelity workflows automate other parts of the process: EL MFS combines low fidelity models inside hierarchical fusion (Wang et al., 2024), AQBMF screens low fidelity information and surrogate candidates (Lee et al., 2026), and the MAESTRO

preprint screens families by leave one out validation (Kim et al., 2026). Our ensemble fitting stage acts on completed target field predictions from different fidelity strategies. We evaluate that stage and its library, without a direct comparison against these complete engineering automation systems.

The library draws on Fourier neural operators (FNO) (Li et al., 2021), DeepONet (Lu et al., 2021), wavelet neural operators (WNO) (Tripura & Chakraborty, 2022), and Transolver (Wu et al., 2024). Multifidelity operator learning also includes DeepONet (Howard et al., 2023), MF WNO (Thakur et al., 2022), and FNO for carbon storage (Tang et al., 2023). MFRNP uses decoded coarse outputs in residual prediction (Niu et al., 2024), while FIRE uses coarse predictive distributions (Yu et al., 2026). Appendix B.1 identifies the components we adapt and the departures from their original formulations. HyperNOs addresses automated operator training (Ghiotto, 2026), and LANE SI combines sea ice fields (Borisova & Nikitin, 2024). PDEBench and The Well supply broader physical benchmarks (Takamoto et al., 2022, Ohana et al., 2024). Our study starts from supplied training data and fits ensemble weights, leaving simulation acquisition to approaches such as active multi resolution learning (Li et al., 2024).

3 AUTOMF: MODEL SELECTION AND ENSEMBLE FITTING

3.1 FROM PARAMETER INPUTS TO FIELD PREDICTIONS

At fidelity ℓ , the training data are $\mathcal{D}_\ell = \{(x_i^{(\ell)}, y_i^{(\ell)})\}_{i=1}^{N_\ell}$: N_ℓ examples indexed by i , each pairing d simulation parameters $x_i^{(\ell)} \in \mathbb{R}^d$ with a field $y_i^{(\ell)}$ on grid Ω_ℓ . The index ℓ identifies fidelity. Each surrogate model f_m maps the parameter vector x to a fine field on a common target grid. For models that incorporate a coarse field, this field is obtained internally: either a trained low fidelity surrogate model predicts it at x , or nearest neighbour retrieval selects the stored coarse solution whose training parameters are closest to x in standardized parameter space. The coarse field then serves as a reference for a learned correction. Thus, f_m denotes the complete inference procedure, including any coarse prediction, retrieval, and correction, and requires no additional simulation at the query input.

We reserve K additional fine examples for *ensemble fitting* and a separate evaluation set. We refer to the reserved input and fine field pairs as *fitting examples*. Both sets are excluded from low fidelity (LF) and high fidelity (HF) training before normalization, basis construction, or model fitting. Input identities establish correspondence between fidelities. Each dataset has its own trained surrogate library and ensemble weights. Each rule determines its weights from the fitting examples and saves them before scoring evaluation fields. Appendix A.2 discusses split integrity and prior access to historical evaluation results.

3.2 SURROGATE MODEL LIBRARY

The surrogate library contains nine models, M1 to M9 in Table 2, chosen to vary both the representation of a field and the way coarse data enter learning. Transfer can reuse features across fidelities, joint supervision shares information during training, and correction uses a coarse field as a spatial reference. A fine only reduced basis model supplies an alternative when coarse information is less useful. We adapt these strategies to the same parameter input and target field interface, including the internal coarse prediction or retrieval required by correctors. Three models first learn from coarse fields and then finetune on fine fields, following multifidelity transfer learning (Lyu et al., 2023). FiLM FNO transfer combines Fourier mixing (Li et al., 2021) with parameter conditioned featurewise linear modulation (FiLM) (Perez et al., 2018). ConvNeXt transfer uses convolutional blocks in an encoder and decoder with skip connections (Liu et al., 2022). Wavelet transfer uses a custom wavelet representation inspired by WNO (Tripura & Chakraborty, 2022). Together they test coarse initialization across different spatial representations. All pairs FNO instead pools supervision across fidelity pairs before fine training. It takes parameters and two fidelity labels as inputs. Poseidon motivates pair enumeration (Herde et al., 2024), while our FNO (Li et al., 2021) uses neither its temporal operator nor a supplied source field.

Four models use coarse predictions or retrieved fields to guide a correction. Distribution FNO predicts a residual from coarse ensemble summaries, adapting FIRE’s distribution conditioning idea (Yu et al., 2026). Retrieved field DeepONet uses a branch and trunk representation (Lu et al., 2021,

2022) to correct a neighboring coarse training field. The slice attention and ConvNeXt correctors refine a predicted coarse field through six updates, using modified Transolver inspired attention (Wu et al., 2024) and convolutional blocks (Liu et al., 2022), respectively. Their coarse inputs are assembled out of fold with four ensemble members per case. Finally, POD GP predicts coefficients in a proper orthogonal decomposition (POD) basis using Gaussian processes (GP) (Higdon et al., 2008). It tests whether a compact field representation can be learned directly from fine solutions. Our adaptations specify how parameters condition each model, how fidelities supervise it, and how coarse references are constructed. Appendix B.1 explains for each model the source components, our changes, and the inclusion rationale. Architecture and training cost both vary across the library, so rankings cannot be attributed to one design choice alone.

3.3 FITTING ONE WEIGHT PER MODEL

The ensemble averages the model predictions,

$$\hat{y}_w(x) = \sum_{m=1}^M w_m f_m(x), \quad w \in \Delta_M = \{w \in \mathbb{R}^M : w_m \geq 0, \mathbf{1}^\top w = 1\}. \quad (1)$$

Here $M = 9$ counts models, w_m weights model m , and $\hat{y}_w$ is the combined field. The simplex Δ_M contains nonnegative weights summing to one. The vector $\mathbf{1}$ contains ones and $^\top$ denotes transpose. Weights stay fixed across inputs and grid points. For fitting pair (x_i, y_i) , define $\langle a, b \rangle_Q = a^\top Q b$, $\|a\|_Q = \sqrt{a^\top Q a}$, and $s_i = \max(\|y_i\|_Q, 10^{-8})$. Fields are vectors of grid values, $Q = I$ is the identity matrix, and the floor prevents division by zero. We compute

$$e_{im} = \frac{f_m(x_i) - y_i}{s_i}, \quad [G_i]_{mn} = \langle e_{im}, e_{in} \rangle_Q, \quad \hat{C} = \frac{1}{K} \sum_{i=1}^K G_i. \quad (2)$$

Here e_{im} is model m 's relative error field on example i , G_i records error products between models m and n , and $\hat{C}$ averages these over K fields. Diagonal entries measure individual squared error. Other entries measure error alignment, retaining information needed to score a mixture.

Selected model. This rule gives weight one to model m^* with the lowest mean relative L^2 error on the fitting examples, $m^* = \arg \min_m K^{-1} \sum_i \sqrt{[G_i]_{mm}}$.

Inverse error mixture. This rule gives more weight to individually accurate models:

$$w_m^{\text{inv}} = \frac{1/\max(\hat{C}_{mm}, \tau)}{\sum_j 1/\max(\hat{C}_{jj}, \tau)}, \quad \tau = 10^{-30}. \quad (3)$$

Here w_m^{inv} is the normalized inverse error weight, j runs over all models, and τ prevents division by zero. The weights sum to one.

Fitted mixture. This rule chooses the weights together by minimizing the error of their combined prediction:

$$\hat{w}^{\text{full}} \in \arg \min_{w \in \Delta_M} w^\top \hat{C} w. \quad (4)$$

Here $\hat{w}^{\text{full}}$ minimizes the mean squared relative error $w^\top \hat{C} w$ on the fitting examples. This is convex stacking (Wolpert, 1992, Breiman, 1996), allowing a less accurate model to offset another's errors. Both mixture rules use squared normalized errors, whereas model selection and reporting use mean relative L^2 . The objectives can have different optima. We test the consequence of this choice in Section 5 and derive the distinction in Appendix F.1. Numerical details appear in Appendix C.1.

Ensemble fitting estimates only nine weights, with eight free values because they sum to one. The surrogate models remain fixed, and each fitting field supplies spatial error products that summarize individual accuracy and error alignment. Once these products are computed, the optimization uses a 9×9 matrix regardless of the number of grid points. This small fitting problem makes limited additional fine data a practical option. Spatial values within a field remain dependent, however, so field resolution does not count as additional independent fitting examples or guarantee generalization from five fields. Appendix C.2 gives the pseudocode.

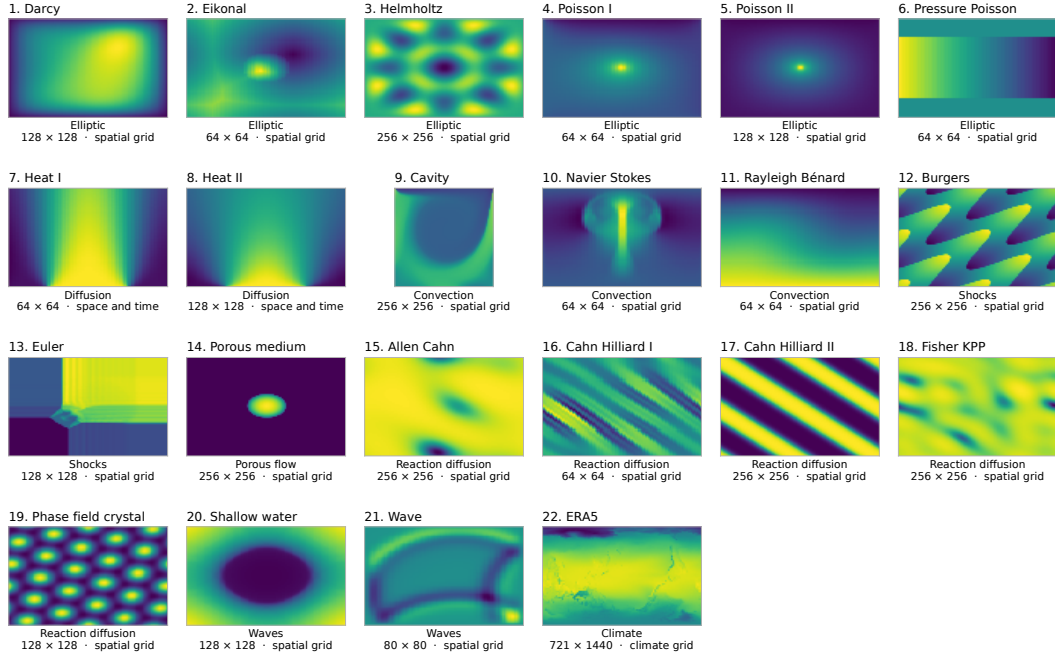


Figure 2: **The 22 datasets span different field structures.** Panels are grouped by class and show one fine training field per dataset with its native grid size. Heat fields span space and time, and the cavity shows vorticity. ERA5 predictions use a 128×256 working grid. Display conventions are described in Appendix A.3. Imported data: Poisson II, Heat II, and Navier Stokes (Niu et al., 2024), and ERA5 (Hersbach et al., 2020, Niu et al., 2024).

4 EVALUATING THE SURROGATE LIBRARY AND ENSEMBLE

4.1 DATASETS AND COMPARISON COVERAGE

We assemble a benchmark of 22 datasets spanning elliptic equations, diffusion, convection, shocks, porous flow, reaction diffusion, waves, and ERA5 climate emulation (Hersbach et al., 2020, Niu et al., 2024). Figure 2 shows one fine field from each dataset, illustrating the range of field structures and grid resolutions. The collection combines our simulations with imported parameter and field data, using different grids or data sources as fidelities. Euler, Burgers, and Shallow water use the published PyClaw finite volume framework (Ketcheson et al., 2012). Their problem designs include Euler quadrant configurations motivated by The Well (Ohana et al., 2024) and a radial dam break adapted from PDEBench (Takamoto et al., 2022). Custom solvers are supported by targeted analytical and numerical checks, including a Barenblatt solution test for Porous medium and mesh refinement for Allen Cahn (Appendix A.1). Several datasets describe the same governing equation, so the roster represents 22 prediction tasks rather than 22 independent physical families (Appendix A).

We evaluate all nine library models and their ensembles on all 22 datasets. Models are trained using the training examples available for each dataset, with additional fine examples reserved for ensemble fitting. Reported comparisons use the same evaluation cases within each dataset and fitting partition. Each final surrogate model uses a single seed. Baseline settings, model specific training schedules, fitting and validation allocations, and comparison coverage are detailed in Appendices B and D.1.

4.2 ENSEMBLE FITTING BUDGETS AND EVALUATION

For each PDE dataset, we keep the trained models fixed and ask how many additional fine solutions are needed to choose their mixture. We compare $K \in \{5, 10, 20\}$ fitting examples, each consisting of an input and its known fine field. These examples are separate from surrogate training and evaluation. Every rule receives the same K examples for selecting an individual model or fitting

mixture weights. We then measure prediction error on separate evaluation cases shared by all methods. We repeat this comparison using three partitions of fitting and evaluation cases without retraining the models. This measures sensitivity to the choice of fitting examples. Split construction and case counts are given in Appendix A.2.

We report mean relative L^2 error, averaging over cases and then partitions. To compare proportional changes across datasets, we aggregate error ratios geometrically. Dataset aggregation gives each dataset equal weight. PDE class aggregation first averages within each class and then gives the classes equal weight:

$$\rho_{a/b} = \exp \left[\frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathcal{D}_c|} \sum_{d \in \mathcal{D}_c} \log \frac{\bar{E}_{d,a}}{\bar{E}_{d,b}} \right]. \quad (5)$$

Here a, b denote methods, $\bar{E}_{d,a}$ is method a 's partition averaged error on dataset d , $\mathcal{C}$ is the set of PDE classes, and $\mathcal{D}_c$ contains class c 's datasets. Vertical bars count set members. The ratio $\rho_{a/b}$ gives each class equal weight, with values below one favoring a . The two summaries answer different questions: a dataset in a singleton class has six times the weight of one elliptic dataset under class aggregation. Wins and losses require more than a 1% relative change. This is a descriptive threshold, not a significance test. Tables express errors as percentages and improvements as relative changes.

5 RESULTS: FROM A SURROGATE LIBRARY TO A FITTED ENSEMBLE

5.1 DIFFERENT MODELS LEAD ON DIFFERENT DATASETS

The library contains strong individual models, but their relative accuracy depends on the prediction task. Seven of its nine members achieve the lowest error on at least one PDE dataset. Wavelet transfer leads on Darcy, POD GP on the Poisson variants, and distribution FNO on the heat variants, while the ConvNeXt corrector leads on Cavity and Euler. The four additional reaction diffusion tasks each favor a different library member. This variation motivates keeping several ways of using coarse information available: choosing one architecture for the entire collection would discard models that are useful on particular problems. Table 1 shows these differences directly, placing each individual model beside the ensemble rules and the strongest additional baseline for each dataset.

The best library member outperforms the best available additional baseline on 15 of the 21 PDE datasets. Here the best model in each collection is identified after evaluation, so this comparison measures the accuracy available within the library. The fitted mixture also achieves lower aggregate error than the best additional baseline, while individual baselines remain preferable on some datasets (Appendix D.2). The next question is whether the reserved fitting examples can turn this available accuracy into a useful prediction rule. Table 2 summarizes the aggregate comparisons and Elo ranking, with individual baseline errors in Appendix D.1.

5.2 COMBINING PREDICTIONS OFTEN IMPROVES ACCURACY

Using the same five fitting examples, the fitted mixture improves on the Selected model on 16 of 22 datasets. It reduces geometric mean relative L^2 error by 11.6% across PDE classes and 7.5% across all datasets. Both rules make their decisions before observing the evaluation fields, so this comparison tests how effectively they use the available fine solutions. The advantage over a best individual model identified retrospectively from evaluation answers is much smaller and depends on aggregation (Appendix D.2). The practical benefit is therefore lower error when model choice must be made before the next fine field is known, with failures as well as improvements visible in Table 1. Figure 3 provides one field example.

5.3 SIMPLE WEIGHTING CAPTURES MOST OF THE BENEFIT

The inverse error mixture nearly matches the fitted mixture in aggregate accuracy, and the two occupy the highest Elo positions in Table 2. Uniform averaging performs poorly (Table 7), indicating that which models receive weight matters. Yet individual model errors already provide a useful guide to those weights, with only a small additional gain from joint fitting. The error analysis supports complementary predictions as a source of improvement (Appendix F.1). Both weighting rules can

Table 1: **Comparing the library, its combinations, and additional baselines.** All columns report mean relative L^2 error in percent on the same evaluation cases. The best additional baseline is identified by its B label, and M1 to M9 are defined in Table 2. The selected model uses one member, the inverse error mixture weights individual errors, and the fitted mixture optimizes their combination. All three use five fitting examples. Bold marks the lowest unrounded evaluation error. Daggers mark exclusions in the separate sensitivity analysis (Appendix E.1). Section 4 specifies coverage. Imported data: Poisson II, Heat II, and Navier Stokes (Niu et al., 2024), and ERA5 (Hersbach et al., 2020, Niu et al., 2024).

Dataset	Best baseline	M1	M2	M3	M4	M5	M6	M7	M8	M9	Selected model	Inverse error mixture	Fitted mixture
Darcy	3.37 (B1)	3.07	3.44	2.78	4.98	2.32	31.7	3.37	3.41	3.41	2.52	2.55	2.33
Eikonal	2.21 (B8)	2.42	2.15	1.81	4.79	2.59	7.37	3.63	3.74	2.95	1.81	1.76	1.76
Helmholtz	4.12 (B9)	3.3	3.48	4.79	10.9	5.04	29.1	7.1	10.9	10.8	6.25	6.89	6.55
Poisson I	0.00531 (B2)	0.0413	0.0387	0.0427	0.0611	0.0955	9.92	0.00684	0.128	0.0838	0.00684	0.008	0.00788
Poisson II [†]	0.0128 (B1)	0.256	0.536	0.2	25.5	0.881	26.6	0.0121	0.921	0.798	0.0121	0.0125	0.0129
Pressure Poisson [†]	0.0639 (B9)	0.0599	0.0695	0.0918	0.0871	0.0498	0.285	0.0922	0.309	0.145	0.0734	0.0769	0.0802
Heat I	0.0206 (B10)	0.0164	0.0125	0.0105	0.00915	0.0268	2.28	0.327	0.0494	0.0149	0.00972	0.00593	0.0062
Heat II	0.0201 (B10)	0.0139	0.00878	0.00859	0.00662	0.0173	1.58	0.312	0.0268	0.0121	0.00662	0.00437	0.00435
Cavity	0.0419 (B5)	98.5	0.29	98.5	0.681	3.6	70.1	0.409	2.06	0.164	0.164	0.142	0.142
Navier Stokes	1.52 (B10)	1.55	1.69	1.57	1.58	2.33	4.41	2.79	2.62	1.95	1.61	1.52	1.5
Rayleigh Bénard	0.011 (B10)	0.0112	0.00915	0.00715	0.00301	0.0137	0.0999	0.295	0.0529	0.0195	0.00301	0.00286	0.00269
Burgers	0.562 (B9)	0.547	0.533	100	0.507	1.28	4.88	2.5	1.69	0.545	0.516	0.513	0.502
Euler	5.54 (B2)	6.2	6.08	6.72	6.3	7.43	8.13	6.15	6	5.88	6.09	5.64	5.73
Porous medium	0.0718 (B9)	0.0448	0.0325	0.0596	0.0654	0.126	2.75	0.909	0.681	0.0497	0.0325	0.0262	0.0263
Allen Cahn	41.2 (B5)	53	52.7	42.1	45.4	41.3	102	55.5	41.4	38.8	39.4	37.9	34.6
Cahn Hilliard I [†]	13.3 (B10)	12.9	12.5	12.2	11.7	36.3	81.6	30.3	41.1	12.5	12.2	11.3	10.7
Cahn Hilliard II	49.8 (B5)	46.9	48.3	51.9	50.8	60.8	99.8	64.8	55.9	47.8	52.2	47.2	49.3
Fisher KPP	3.24 (B9)	3.21	3.22	2.36	2.98	3.83	6.49	6.86	2.86	2.84	2.36	2.53	2.3
Phase field crystal	83.5 (B11)	89.6	89.5	111	83.3	202	120	83.5	85.3	85.7	83.5	83.8	83.3
Shallow water	0.403 (B10)	0.428	0.367	0.362	0.876	0.446	1.38	1.09	1.29	0.502	0.367	0.357	0.389
Wave	3.3 (B3)	6.36	6.31	4.75	11.1	5.04	64.2	24.9	8.35	11.3	4.84	4.39	4.27
ERA5	5 (B1)	14.2	17.7	20.4	12.2	9.75	9.54	6.74	20.4	8.24	6.74	5.81	6.49

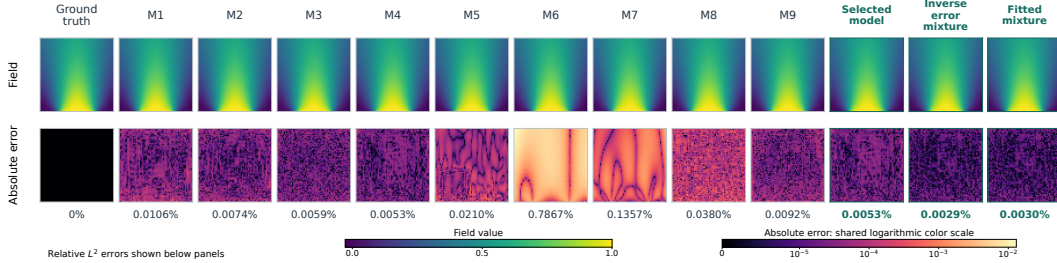


Figure 3: **Combining models reduces field error on a Heat I example.** Top: ground truth, nine model predictions, and the selected model, inverse error mixture, and fitted mixture. Bottom: absolute error on a shared logarithmic color scale, with relative L^2 error below each panel. Fields share a linear color scale and a 64×64 space and time grid. The three rules use five fitting examples. The selected model is M4 here. The case is the first evaluation input of the first partition. The fitted mixture lowers error by 44.4% relative to the best individual model.

benefit from that complementarity, even though the inverse error mixture does not explicitly estimate interactions between model errors.

5.4 SENSITIVITY TO FITTING AND EVALUATION CHOICES

The advantage over model selection persists when more fine examples are available for fitting. Across the 21 PDE datasets, increasing the allowance from five to twenty examples raises the fitted mixture’s reduction in class aggregated geometric mean relative L^2 error from 11.6% to 13.1% (Appendix D.3). Both methods receive the same allowance at each budget, and the surrogate models remain fixed. The modest increase supports using a small fitting set in these experiments, while leaving the amount needed for a new problem to be assessed.

Table 2: **Model sources and performance, ranked by Elo.** For each dataset, ratios divide a method’s mean evaluation error by the Selected model’s error before geometric averaging. The Selected model is the member of M1 to M9 with the lowest mean relative L^2 error on the five fitting examples. **PDE class ratio** averages within each class and then equally across the seven PDE classes (21 datasets). **Dataset ratio** weights all 22 datasets equally, including ERA5. A ratio of 1 matches selection, 0.90 means 10% lower aggregate error, and 1.10 means 10% higher aggregate error. Elo compares each pair of methods on all 22 datasets, with higher ratings better (Appendix D.4). M1 to M9 form the library, and B1 to B11 are additional baselines. Citations identify components or ideas, with adaptations in Appendix B.1. Individual errors appear in Table 1 and Appendix D.1. Training costs and fine data budgets differ.

Model / rule	Component or idea source	PDE class ratio	Dataset ratio	Elo
Fitted mixture	(Breiman, 1996)	0.884	0.925	1988
Inverse error mixture	(Goel et al., 2007)	0.888	0.932	1951
Selected model	Fitting set selection	1.000	1.000	1851
M2 All pairs FNO	(Li et al., 2021, Herde et al., 2024)	1.333	1.541	1688
M1 FiLM FNO transfer	(Li et al., 2021, Perez et al., 2018)	1.940	2.035	1642
M9 ConvNeXt corrector	(Liu et al., 2022)	1.695	1.867	1634
M3 ConvNeXt transfer	(Liu et al., 2022)	2.607	2.445	1619
M4 Distribution FNO	(Yu et al., 2026)	1.795	2.155	1592
B10 FNO transfer II	(Lyu et al., 2023)	1.794	1.850	1590
B9 FNO transfer I	(Lyu et al., 2023)	1.794	1.848	1578
M5 Wavelet transfer	(Tripura & Chakraborty, 2022)	2.459	2.400	1528
B5 Latent MF network	(Li et al., 2022a)	3.794	2.772	1497
M7 POD GP	(Higdon et al., 2008)	5.492	3.108	1450
B1 Autoregressive POD GP	(Kennedy & O’Hagan, 2000)	5.854	3.362	1449
M8 Slice attention corrector	(Wu et al., 2024)	4.189	3.549	1439
B2 Nonlinear POD GP	(Perdikaris et al., 2017)	5.871	3.626	1424
B3 Composite MF MLP	(Meng & Karniadakis, 2020)	4.174	3.974	1387
B8 Fidelity basis FNO	(Li et al., 2022b)	8.979	6.217	1364
B4 Composite MF DeepONet	(Howard et al., 2023)	8.001	5.803	1337
B11 Affine POD GP	(Zhang et al., 2018)	17.624	11.236	1218
B6 Nonlinear decoder transfer	(Seidman et al., 2022)	9.927	8.252	1197
M6 Retrieved field DeepONet	(Lu et al., 2022)	20.847	14.890	1057
B7 MFRNP adapter	(Niu et al., 2024)	49.969	24.194	1020

Two diagnostics use the same 17 dataset subset to examine dependence on the evaluation design. Repeating the comparisons on fixed dataset subsets preserves the gain over the Selected model, while superiority to the best individual model remains sensitive to the comparison (Appendix E.1). We also minimize the reported mean relative L^2 error directly, keeping the trained models and fitting examples fixed. This gives a modest improvement over fitting squared relative error (Table 12), identifying the fitting objective as a useful design choice. This retrospective control is reported separately from the three main rules.

6 ANALYSIS: WHY AVERAGING CAN HELP

Averaging can improve predictions when spatial errors partly offset. For a fixed library, classical ensemble analysis writes the expected squared relative error as

$$R(w) = \mathbb{E}[w^\top G(x, y)w] = w^\top Cw, \quad C = \mathbb{E}[G(x, y)]. \quad (6)$$

Here $G(x, y)$ is the error product matrix for a new input and target pair, $\mathbb{E}$ averages over such pairs, C is its population mean, and $R(w)$ is the mixture’s expected squared relative error. Its diagonal measures individual error and its other entries retain alignment and shared bias (Breiman, 1996, Krogh & Vedelsby, 1994). The inverse error mixture can exploit complementary predictions even though it uses only individual accuracy to choose weights.

This identity does not guarantee useful weights from a small fitting set, and the bound in Appendix F.2 does not certify accuracy from five fields. The reported metric $L(w) = \mathbb{E}[\sqrt{w^\top G w}]$ averages unsquared relative error and can favor different weights from R even with unlimited fitting data. Appendix F.1 gives a counterexample and a direct fitting control.

7 APPLICATION TO CLIMATE FIELD EMULATION

ERA5 asks whether the workflow remains useful beyond the simulation datasets. The task maps 12 emissions related inputs to temperature fields, combining MFRNP’s climate assembly (Niu et al., 2024) with ERA5 reanalysis (Hersbach et al., 2020), in the climate emulation setting motivated by ClimateBench (Watson-Parris et al., 2022). It contains 55 fine training inputs, ten fitting inputs, and seven evaluation inputs. Reserved inputs are excluded from all training fidelities. All nine models predict on a 128×256 working grid.

POD GP is the most accurate of the nine library models on ERA5, with a mean relative L^2 error of 6.7369%. With five fitting examples, the inverse error mixture reduces this error to 5.8146%, a 13.69% relative improvement. The fitted mixture reaches 6.4877%, improving by 3.70%. Both mixtures therefore improve on the best individual model in the library. Table 1 shows the individual models and mixtures alongside the strongest additional baseline, with all baseline results in Appendix D.1.

8 DISCUSSION AND LIMITATIONS

The findings connect the diversity of the surrogate library to a practical model choice procedure. Different representations and uses of coarse data provide accurate predictions on different problems. Reserved fine examples then guide how much each model contributes before the next field is known. Mixtures often improve on selecting one model, while inverse error weighting shows that much of this gain is accessible through individual error estimates. For an application with expensive simulations, additional surrogate computation can therefore be a useful way to exploit the available fields. Once the library is trained, comparing and fitting the three rules reuses its predictions without retraining the models.

These results do not isolate a benefit specific to multifidelity data. The library includes a fine only model, many datasets have equally sized coarse and fine training tables, and validation allocations vary. Establishing data efficiency requires a fine only library and alternative uses of fitting labels under equal total label and computation budgets, counting every candidate and shared coarse ensemble. Whether the additional surrogate computation is worthwhile depends on simulation cost, the number of prediction queries, and the required accuracy. Net cost savings and comparison with complete engineering automation systems remain open.

9 CONCLUSION

The strongest surrogate for multifidelity field prediction changes across problems. AUTOMF combines an adapted library of nine neural and reduced basis models with an ensemble fitting workflow. Reserved fine solutions determine which models to select or combine, and new inputs require no additional coarse simulation. The assembled benchmark compares these choices across field structures and fidelity relationships.

Across 22 datasets, the fitted mixture improves on selecting one model on 16 datasets using five reserved fine examples. Inverse error weighting captures most of this benefit. ERA5 extends the comparison beyond PDE simulations. These findings support ensemble fitting when uncertain model choice and expensive field data justify additional surrogate computation.

REFERENCES

- E. Acar and M. Rais-Rohani. Ensemble of metamodels with optimized weight factors. *Structural and Multidisciplinary Optimization*, 37:279–294, 2009. doi: 10.1007/s00158-008-0230-y. URL <https://doi.org/10.1007/s00158-008-0230-y>.
- Erdem Acar. Effect of error metrics on optimum weight factor selection for ensemble of metamodels. *Expert Systems with Applications*, 42(5):2703–2709, 2015. doi: 10.1016/j.eswa.2014.11.020. URL <https://doi.org/10.1016/j.eswa.2014.11.020>.
- Julia Borisova and Nikolay O. Nikitin. Surrogate modelling for sea ice concentration using lightweight neural ensemble. In *AAAI Workshop on AI to Accelerate Science and Engineering*, 2024. URL <https://arxiv.org/abs/2312.04330>. Preprint arXiv:2312.04330, 2023.
- Leo Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996. doi: 10.1023/A:1018046112532. URL <https://statistics.berkeley.edu/sites/default/files/tech-reports/367.pdf>.
- Lucas Brunel, Mathieu Balesdent, Loïc Brévault, Rodolphe Le Riche, and Bruno Sudret. A survey on multi-fidelity surrogates for simulators with functional outputs: unified framework and benchmark. *Computer Methods in Applied Mechanics and Engineering*, 435:117577, 2025. doi: 10.1016/j.cma.2024.117577. URL <https://doi.org/10.1016/j.cma.2024.117577>.
- Rich Caruana, Alexandru Niculescu-Mizil, Geoff Crew, and Alex Ksikes. Ensemble selection from libraries of models. In *Proceedings of the 21st International Conference on Machine Learning*, 2004. doi: 10.1145/1015330.1015432. URL <https://doi.org/10.1145/1015330.1015432>.
- K. R. Elder and Martin Grant. Modeling elastic and plastic deformations in nonequilibrium processing using phase field crystals. *Physical Review E*, 70:051605, 2004. doi: 10.1103/PhysRevE.70.051605. URL <https://doi.org/10.1103/PhysRevE.70.051605>.
- Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola. AutoGluon-Tabular: Robust and accurate AutoML for structured data. In *7th ICML Workshop on Automated Machine Learning*, 2020. URL <https://arxiv.org/abs/2003.06505>.
- Ghifari Adam Faza, Kemas Zakaria, Pramudita Satria Palar, Keivan Shariatmadar, Lavi Rizki Zuhail, Hans Hallez, and David Moens. Benchmarking multi-fidelity neural operators on complex PDE problems with non-trivial fidelity differences, 2026. URL <https://arxiv.org/abs/2608.04708>.
- Matthias Feurer, Aaron Klein, Katharina Eggensperger, Jost Springenberg, Manuel Blum, and Frank Hutter. Efficient and robust automated machine learning. In *Advances in Neural Information Processing Systems*, volume 28, 2015. URL https://papers.nips.cc/paper_files/paper/2015/hash/11d0e6287202fcd83f79975ec59a3a6-Abstract.html.
- Matthias Feurer, Katharina Eggensperger, Stefan Falkner, Marius Lindauer, and Frank Hutter. Auto-Sklearn 2.0: Hands-free AutoML via meta-learning. *Journal of Machine Learning Research*, 23, 2022. URL <https://jmlr.org/papers/v23/21-0992.html>.
- Massimiliano Ghitto. HyperNOs: automated and parallel library for neural operators research. *Bollettino dell’Unione Matematica Italiana*, 19:709–743, 2026. doi: 10.1007/s40574-025-00516-0. URL <https://doi.org/10.1007/s40574-025-00516-0>. First published online October 21, 2025.
- Tushar Goel, Raphael T. Haftka, Wei Shyy, and Nestor V. Queipo. Ensemble of surrogates. *Structural and Multidisciplinary Optimization*, 33:199–216, 2007. doi: 10.1007/s00158-006-0051-9. URL <https://doi.org/10.1007/s00158-006-0051-9>.
- Maximilian Herde, Bogdan Raonić, Tobias Rohner, Roger Käppeli, Roberto Molinaro, Emmanuel de Bézenac, and Siddhartha Mishra. Poseidon: Efficient foundation models for PDEs. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/84e1blec17bb11c57234e96433022a9a-Abstract-Conference.html.

- Hans Hersbach, Bill Bell, Paul Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146:1999–2049, 2020. doi: 10.1002/qj.3803. URL <https://doi.org/10.1002/qj.3803>.
- Dave Higdon, James Gattiker, Brian Williams, and Maria Rightley. Computer model calibration using high-dimensional output. *Journal of the American Statistical Association*, 103(482): 570–583, 2008. doi: 10.1198/016214507000000888.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830. URL <https://doi.org/10.1080/01621459.1963.10500830>.
- Amanda A. Howard, Mauro Perego, George E. Karniadakis, and Panos Stinis. Multifidelity deep operator networks for data-driven and physics-informed problems. *Journal of Computational Physics*, pp. 112462, 2023. doi: 10.1016/j.jcp.2023.112462. URL <https://arxiv.org/abs/2204.09157>.
- Marc C. Kennedy and Anthony O’Hagan. Predicting the output from a complex computer code when fast approximations are available. *Biometrika*, 87(1):1–13, 2000. doi: 10.1093/biomet/87.1.1. URL <https://doi.org/10.1093/biomet/87.1.1>.
- David I. Ketcheson, Kyle T. Mandli, Aron J. Ahmadi, Amal Alghamdi, Manuel Quezada de Luna, Matteo Parsani, Matthew G. Knepley, and Matthew Emmett. PyClaw: Accessible, extensible, scalable tools for wave propagation problems. *SIAM Journal on Scientific Computing*, 34(4):C210–C231, 2012. doi: 10.1137/110856976. URL <https://doi.org/10.1137/110856976>.
- Jieon Kim, Taeyeon Park, Sanghyun Cho, Jieun Park, and Gunwoo Noh. MAESTRO: A multi-fidelity modeling assisted by an ensemble of surrogates for robust prediction. SSRN preprint, 2026. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6207840. Posted February 10, 2026.
- Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems*, volume 7, 1994. URL <https://proceedings.neurips.cc/paper/1994/hash/b8c37e33defde51cf91e03e51657da-Abstract.html>.
- Mingyu Lee, Juyoung Lee, Jae-Hoon Choi, Nam H. Kim, and Ikjin Lee. A novel adaptive quality-based multi-fidelity surrogate framework for multiple low-fidelity data sources. *Advanced Engineering Informatics*, 69:103973, 2026. doi: 10.1016/j.aei.2025.103973. URL <https://web.mae.ufl.edu/nkim/Papers/paper151.pdf>. Available online October 17, 2025.
- Shibo Li, Zheng Wang, Robert M. Kirby, and Shandian Zhe. Deep multi-fidelity active learning of high-dimensional outputs. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pp. 1694–1711, 2022a. URL <https://proceedings.mlr.press/v151/li22b.html>.
- Shibo Li, Zheng Wang, Robert M. Kirby, and Shandian Zhe. Infinite-fidelity coregionalization for physical simulation. In *Advances in Neural Information Processing Systems*, volume 35, 2022b. URL <https://arxiv.org/abs/2207.00678>.
- Shibo Li, Xin Yu, Wei Xing, Robert Kirby, Akil Narayan, and Shandian Zhe. Multi-resolution active learning of fourier neural operators. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pp. 2440–2448, 2024. URL <https://proceedings.mlr.press/v238/li24k.html>.
- Zongyi Li, Nikola Kovachki, Kamyar Aizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2010.08895>.

- Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Liu_A_ConvNet_for_the_2020s_CVPR_2022_paper.html.
- Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021. doi: 10.1038/s42256-021-00302-5. URL <https://www.nature.com/articles/s42256-021-00302-5>.
- Lu Lu, Raphaël Pestourie, Steven G. Johnson, and Giuseppe Romano. Multifidelity deep neural operators for efficient learning of partial differential equations with application to fast inverse design of nanoscale heat transport. *Physical Review Research*, 4(2):023210, 2022. doi: 10.1103/PhysRevResearch.4.023210.
- Yanfang Lyu, Xiaoyu Zhao, Zhiqiang Gong, Xiao Kang, and Wen Yao. Multi-fidelity prediction of fluid flow and temperature field based on transfer learning using Fourier Neural Operator, 2023. URL <https://arxiv.org/abs/2304.06972>.
- Xuhui Meng and George Em Karniadakis. A composite neural network that learns from multi-fidelity data: Application to function approximation and inverse PDE problems. *Journal of Computational Physics*, 401:109020, 2020. doi: 10.1016/j.jcp.2019.109020.
- Ruijia Niu, Dongxia Wu, Kai Kim, Yian Ma, Duncan Watson-Parris, and Rose Yu. Multi-fidelity residual neural processes for scalable surrogate modeling. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 38381–38394, 2024. URL <https://proceedings.mlr.press/v235/niu24d.html>.
- Ruben Ohana, Michael McCabe, Lucas Meyer, Rudy Morel, Fruzsina J. Agocs, et al. The well: a large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/4f9a5acd91ac76569f2fe291b1f4772b-Abstract-Datasets_and_Benchmarks_Track.html.
- Lauren Partin, Gianluca Geraci, Ahmad Rushdi, Michael S. Eldred, and Daniele E. Schiavazzi. Multifidelity data fusion in convolutional encoder/decoder networks. *Journal of Computational Physics*, 2022. doi: 10.1016/j.jcp.2022.111666. URL <https://arxiv.org/abs/2205.05187>.
- Paris Perdikaris, Maziar Raissi, Andreas Damianou, Neil D. Lawrence, and George E. Karniadakis. Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. *Proceedings of the Royal Society A*, 473(2198):20160751, 2017. doi: 10.1098/rspa.2016.0751.
- Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. URL <https://ojs.aaai.org/index.php/AAAI/article/view/11671>.
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention*, pp. 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28. URL <https://arxiv.org/abs/1505.04597>.
- Jacob H. Seidman, Georgios Kissas, Paris Perdikaris, and George J. Pappas. NOMAD: Nonlinear manifold decoders for operator learning. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/24f49b2ad9fbc65eefbfd99d6f6c3fd2-Abstract-Conference.html.
- Ira J. S. Shokar, Rich R. Kerswell, and Peter H. Haynes. Conditioning on PDE parameters to generalise deep learning emulation of stochastic and chaotic dynamics. arXiv preprint arXiv:2509.09599, 2025. URL <https://arxiv.org/abs/2509.09599>.

- Niclas Strömberg. Comparison of optimal linear, affine and convex combinations of metamodels. *Engineering Optimization*, 53(4):702–718, 2021. doi: 10.1080/0305215X.2020.1746781. URL <https://doi.org/10.1080/0305215X.2020.1746781>.
- Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Dan MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/0a9747136d411fb83f0cf81820d44afb-Abstract-Datasets_and_Benchmarks.html.
- Hewei Tang, Qingkai Kong, and Joseph P. Morris. Multi-fidelity fourier neural operator for fast modeling of large-scale geological carbon storage, 2023. URL <https://arxiv.org/abs/2308.09113>.
- Akshay Thakur, Tapas Tripura, and Souvik Chakraborty. Multi-fidelity wavelet neural operator with application to uncertainty quantification, 2022. URL <https://arxiv.org/abs/2208.05606>.
- Tapas Tripura and Souvik Chakraborty. Wavelet neural operator: a neural operator for parametric partial differential equations, 2022. URL <https://arxiv.org/abs/2205.02191>.
- Mark J. van der Laan, Eric C. Polley, and Aaron E. Hubbard. Super learner. *Statistical Applications in Genetics and Molecular Biology*, 6(1):Article 25, 2007. doi: 10.2202/1544-6115.1309. URL <https://doi.org/10.2202/1544-6115.1309>.
- Juan Luis Vázquez. *The Porous Medium Equation: Mathematical Theory*. Oxford Mathematical Monographs. Oxford University Press, 2007. doi: 10.1093/acprof:oso/9780198569039.001.0001. URL <https://academic.oup.com/book/7922>.
- Felipe A. C. Viana, Raphael T. Haftka, and Valder Steffen, Jr. Multiple surrogates: how cross-validation errors can help us to obtain the best predictor. *Structural and Multidisciplinary Optimization*, 39:439–457, 2009. doi: 10.1007/s00158-008-0338-0. URL <https://doi.org/10.1007/s00158-008-0338-0>.
- Yitang Wang, Yong Pang, Tianhang Xue, Shuai Zhang, and Xueguan Song. Ensemble learning based hierarchical surrogate model for multi-fidelity information fusion. *Advanced Engineering Informatics*, 60:102535, 2024. doi: 10.1016/j.aei.2024.102535. URL <https://doi.org/10.1016/j.aei.2024.102535>.
- D. Watson-Parris, Y. Rao, D. Olivié, Ø. Seland, P. Nowack, G. Camps-Valls, P. Stier, S. Bouabid, M. Dewey, E. Fons, J. Gonzalez, P. Harder, K. Jeggle, J. Lenhardt, P. Manshausen, M. Novitasari, L. Ricard, and C. Roesch. ClimateBench v1.0: A benchmark for data-driven climate projections. *Journal of Advances in Modeling Earth Systems*, 14:e2021MS002954, 2022. doi: 10.1029/2021MS002954. URL <https://doi.org/10.1029/2021MS002954>.
- Laura J. Wilcox, Robert J. Allen, Bjørn H. Samset, et al. The regional aerosol model intercomparison project (RAMIP). *Geoscientific Model Development*, 16(15):4451–4479, 2023. doi: 10.5194/gmd-16-4451-2023. URL <https://gmd.copernicus.org/articles/16/4451/2023/gmd-16-4451-2023.html>.
- David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992. doi: 10.1016/S0893-6080(05)80023-1. URL [https://doi.org/10.1016/S0893-6080\(05\)80023-1](https://doi.org/10.1016/S0893-6080(05)80023-1).
- Haixu Wu, Huakun Luo, Haowen Wang, Jianmin Wang, and Mingsheng Long. Transolver: A fast transformer solver for PDEs on general geometries. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 53681–53705, 2024. URL <https://proceedings.mlr.press/v235/wu24r.html>.
- Rosen Ting-Ying Yu, Nicholas Sung, and Faez Ahmed. FIRE: Multi-fidelity regression with distribution-conditioned in-context learning using tabular foundation models, 2026. URL <https://arxiv.org/abs/2601.22371>. Also presented at the ICLR 2026 FM4Science workshop.

Yiming Zhang, Nam-Ho Kim, Chanyoung Park, and Raphael T. Haftka. Multifidelity surrogate based on single linear regression. *AIAA Journal*, 56(12):4944–4952, 2018. doi: 10.2514/1.J057299.

A DATASETS AND EVALUATION PROTOCOL

A.1 DATASET DEFINITIONS

Table 3 describes the 22 prediction tasks. Here d counts input parameters, N_H counts HF training rows, and $N_{L,\min}$ is the smallest LF table, not the sum over fidelities. Query inputs are excluded from every training fidelity and are subsequently partitioned into fitting and evaluation sets. These counts describe available data, rather than identical exposure under every training recipe. The seven PDE classes contain six elliptic, two diffusion, three convection, two shock, one porous flow, five reaction diffusion, and two wave datasets. ERA5 supplies the climate case and enters dataset aggregation, but not PDE class aggregation.

Table 3: **Dataset definitions and available training data.** Query counts include fitting pools and evaluation cases. Work grids are used for prediction and scoring. Citations identify imported data, while uncited entries use our simulation code.

Dataset	d	Levels	$N_{L,\min}$	N_H	Query	Work grid	Class
Darcy	16	3	400	400	100	128×128	Elliptic
Eikonal	3	2	400	400	100	64×64	Elliptic
Helmholtz	2	3	400	400	100	256×256	Elliptic
Poisson I	5	3	400	400	100	64×64	Elliptic
Poisson II (Niu et al., 2024)	5	5	64	64	512	128×128	Elliptic
Pressure Poisson	2	4	400	400	100	64×64	Elliptic
Heat I	3	3	400	400	100	64×64	Diffusion
Heat II (Niu et al., 2024)	3	5	1024	1024	512	128×128	Diffusion
Cavity	1	4	400	400	100	256×256	Convection
Navier Stokes (Niu et al., 2024)	2	2	256	256	256	64×64	Convection
Rayleigh Bénard	1	2	400	400	100	64×64	Convection
Burgers	2	3	400	400	100	256×256	Shocks
Euler	7	3	400	400	100	128×128	Shocks
Porous medium	2	3	400	400	100	256×256	Porous flow
Allen Cahn	19	3	400	400	78	256×256	Reaction diffusion
Cahn Hilliard I	2	2	400	400	100	64×64	Reaction diffusion
Cahn Hilliard II	19	3	400	400	100	256×256	Reaction diffusion
Fisher KPP	50	3	400	400	100	256×256	Reaction diffusion
Phase field crystal	18	3	400	400	100	128×128	Reaction diffusion
Shallow water	2	3	400	400	100	128×128	Waves
Wave	3	2	400	400	100	80×80	Waves
ERA5 (Hersbach et al., 2020, Niu et al., 2024)	12	9	313	55	17	128×256	Climate

Poisson II, Heat II, and Navier Stokes use data released with MFRNP (Niu et al., 2024). ERA5 combines its climate model sources with reanalysis targets (Hersbach et al., 2020, Niu et al., 2024). Other entries were simulated for this collection. Pressure Poisson adapts an analytic construction motivated by Partin et al. (2022), Euler a quadrant setup motivated by The Well (Ohana et al., 2024), and Shallow water a radial dam break recipe motivated by PDEBench (Takamoto et al., 2022). These recipe influences do not make the independently simulated entries copies of those benchmarks. Euler, Burgers, and Shallow water call PyClaw (Ketcheson et al., 2012), using first order Godunov methods at coarse fidelities and SharpClaw WENO reconstruction at the fine fidelity. Our dam break varies both the inner water height and radius on a unit square, with independent solves at each grid. Phase field crystal implements the conserved dynamics of Elder & Grant (2004). Roman numerals distinguish tasks with the same equation. Poisson I and II have related parameterizations, while Cahn Hilliard I and II have 2 and 19 input parameters. Cahn Hilliard II, Allen Cahn, and Fisher KPP use 64×64 , 128×128 , and 256×256 fidelity grids. Phase field crystal uses 32×32 , 64×64 , and 128×128 grids. Their input dimensions are 19, 19, 50, and 18, respectively. Heat fields span space and time, and Helmholtz predicts amplitude normalized fields.

Numerical checks and their scope. Verification scripts test the custom implementations against analytical solutions, conservation laws, and refined numerical solutions. In a one dimensional Barenblatt test (Vázquez, 2007), the Porous medium solver’s relative L^2 error falls from 8.1×10^{-4} at 128 points to 2.6×10^{-5} at 512 points. An Allen Cahn mesh check reduces error from 4.9×10^{-5} at 64 points to 2.3×10^{-6} at 256 points against a 512 point reference. The archived checks also test mass conservation for Cahn Hilliard II and Phase field crystal, and the Helmholtz discrete equation

residual. These are controlled checks of selected solver components, including one dimensional tests for solvers used in two dimensions. They do not certify every sampled parameter or the accuracy of every stored field. The companion package records the scripts, measured errors, and scope of each check. Published software and recipe references establish provenance, while the implementation checks are our own.

Related archives can share physical simulations, so the 22 tasks are not independent samples of PDE families. Three entries have additional limitations: Poisson II has inherited numerical defects, Cahn Hilliard I omits realization information from its parameters, and Pressure Poisson uses synthetic coarse fields. Appendix E.1 checks sensitivity to these entries without certifying the other solvers.

A.2 TRAINING, FITTING, AND EVALUATION CASES

Reserved fitting and evaluation inputs are excluded from LF and HF training before normalization, basis construction, or model fitting. Within each PDE partition, input hashes separate fitting from evaluation and define nested budgets $K \in \{5, 10, 20\}$. Every rule uses the same fitting allowance, including any selection or tuning. Models remain fixed across three partitions, which measure sensitivity to case assignment rather than variation across training seeds. Some cases change roles between partitions. Across the 21 PDE datasets, 3,058 query rows produce 1,842 evaluation appearances corresponding to 1,505 distinct dataset and row pairs. Shared simulations across related datasets can further reduce physical independence. Historical evaluation results influenced development and reporting scope, making these comparisons retrospective.

Allen Cahn, Cahn Hilliard II, Fisher KPP, and Phase field crystal each supply 400 training inputs per fidelity. Their query pools contain 78, 100, 100, and 100 inputs. Each partition reserves 16 evaluation cases for Allen Cahn and 20 for each of the other three datasets. ERA5 has 55 HF training inputs, ten fitting pool inputs, and seven evaluation inputs, with a smallest retained LF table of 313 rows. The additional Heat I inputs in Appendix E.3 form a separate supplementary evaluation. The subset and loss diagnostics cover the original 17 PDE tasks, while the primary comparison covers all 22 datasets.

Predictions share target grids, scaling, and input identities within each comparison. Source hashes, saved target identities, or checkpoint evaluation verify correspondence. Saved weights are fixed before evaluation answers are scored, and replay checks reproduce individual errors, mixture errors, and aggregate denominators. ERA5 answer files are separated from model training, and changing evaluation answers leaves fitted weights unchanged in the collector check. These checks verify the implemented separation without reconstructing all prior researcher access. Error matrices use float64 and the same stored model predictions are reused within each experiment. Precision and checkpoint provenance records remain in the companion package.

A.3 HOW THE FIELDS ARE DISPLAYED

Figure 2 shows row zero of each HF training table, chosen independently of prediction error. Its grids are native array grids, while the inventory gives the working grids used for scoring. Heat axes span space and time. The cavity target is vorticity, with the top lid moving right and array rows increasing upward. Its display uses a symmetric logarithmic scale, linear within $[-1, 1]$ and bounded by $[-400, 400]$, so wall values do not hide the interior structure. No values are clipped or modified. Enlarged panels, numerical display scales, extraction metadata, and source hashes for all 22 datasets are included in the companion package.

Figure 3 illustrates Heat I, selected after inspecting dataset results. It uses the first evaluation case of the first partition, not the case with the greatest improvement, and reuses the saved five example fits. All predicted fields share a linear color scale. Absolute errors share a logarithmic scale that is linear between zero and 10^{-5} . Scores use the complete 64×64 fields without smoothing. The selected model is M4, so its column repeats that model. Figure 1’s ERA5 display uses a smoothed coarse thumbnail and interpolates the predicted working grid to the native fine grid for illustration, without changing scored fields.

B MODEL ADAPTATIONS AND TRAINING

The library varies both the representation of a field and how coarse data enter learning. We first identify the retained published components and our adaptations, then give the training settings. The additional baseline descriptions use the same distinction between a source idea and a complete published method.

B.1 MODEL SOURCES, ADAPTATIONS, AND RATIONALE

Each library member combines a field representation with a training and inference procedure. We describe M1 to M9 individually below, identifying the published components retained, the changes made for parameter to field prediction, and the reason for including each model. Several members combine ideas from more than one paper. The resulting library is an integration of adapted models, not nine independent reproductions or nine newly proposed architectures. All accept parameters alone at deployment. Appendix B.2 gives their training budgets and data use. The rationales below motivate the library’s composition, while the experiments assess its predictions. They do not isolate the effect of every architectural change.

M1: FiLM FNO transfer. The starting components are FNO’s learned Fourier mixing (Li et al., 2021), FiLM’s featurewise affine conditioning (Perez et al., 2018), and the LF pretraining followed by HF finetuning strategy of Lyu et al. (2023). We retain spectral convolutions with local residual paths and adapt the input interface to a simulation parameter vector. The spatial input contains coordinates only. At each block, a parameter network generates a scale and shift applied after group normalization, so the parameters influence features throughout the network. Related parameter conditioning for PDE emulation is studied by Shokar et al. (2025). Our model differs from our parameter concatenation FNO references, which supply parameters as constant spatial channels at the input. The same weights are first trained on LF fields and then finetuned on HF fields. This combination tests whether repeated parameter conditioning and a coarse initialization provide an effective way to generate globally coupled fields.

M2: All pairs FNO. M2 retains M1’s FNO and FiLM components (Li et al., 2021, Perez et al., 2018) and adapts the idea of pair enumeration from Poseidon (Herde et al., 2024). Poseidon uses pairs of states for temporal operator learning. Here the pairs index source and target fidelities. We append their normalized labels to the parameter vector, pool the corresponding target fields on a common working grid, and train one conditional FNO before HF finetuning. The source label specifies a training pair, but no source field enters the network. Poseidon’s architecture, pretrained weights, and temporal operator are not used. For ERA5, exact input joins retain each target field with its own parameters. The purpose is to let one model share information across fidelity tasks before specializing to the fine field. Pair enumeration also repeats targets and changes optimization cost, so its effect cannot be attributed to fidelity conditioning alone.

M3: ConvNeXt transfer. ConvNeXt supplies the depthwise spatial convolution, channel expansion, and residual block structure (Liu et al., 2022). We place these blocks in an U shaped encoder and decoder with skip connections (Ronneberger et al., 2015) and a dense field output, adapting the image classification backbone to field generation. The model starts from coordinates and replaces the usual block normalization with group normalization and FiLM conditioning on the simulation parameters (Perez et al., 2018). Three downsampling stages and a bottleneck process the field at several resolutions, with corresponding decoder stages restoring the target grid. LF pretraining and HF finetuning follow the transfer principle of Lyu et al. (2023). The rationale is to provide local spatial processing alongside M1’s Fourier mixing, while retaining the same type of parameter conditioning and transfer schedule.

M4: Distribution FNO. FIRE conditions a fine correction on summaries of a coarse predictive distribution using tabular foundation models and in context inference (Yu et al., 2026). M4 retains the distribution conditioning idea and replaces those models with FNOs trained on spatial fields (Li et al., 2021). Five LF members predict a coarse field at the query parameters. Their pointwise mean, standard deviation, and 10th, 50th, and 90th percentiles become spatial input channels for a residual FNO, together with coordinates and broadcast parameters. The residual is added to the coarse mean

to predict the fine field. Both stages are trained by gradient optimization, so this implementation does not retain FIRE’s inference without model retraining. The ensemble summaries are features of prediction variability, not a guarantee of accurate uncertainty estimates. The rationale is to expose coarse disagreement as well as the mean to the correction model. This recipe has no out of fold construction and is distinct from the coarse ensembles used by M8 and M9.

M5: Wavelet transfer. WNO motivates learning in wavelet coordinates (Tripura & Chakraborty, 2022). We replace M1’s Fourier mixing with a custom single level db4 wavelet transform, while retaining its coordinate input, FiLM conditioning (Perez et al., 2018), local residual paths, and LF to HF transfer schedule (Lyu et al., 2023). Within each wavelet subband, learned weights mix channels at coefficient positions inside a capped leading block. Other positions pass through unchanged before the inverse transform. These indices describe positions within a subband, not a Fourier frequency cutoff. The implementation also handles rectangular grids, odd grid sizes, and one dimensional grids. It is a custom wavelet representation rather than an exact reproduction of the published WNO. Its inclusion tests whether spatially localized wavelet features provide useful predictions alongside the spectral and convolutional transfer models.

M6: Retrieved field DeepONet. DeepONet supplies the branch and trunk representation (Lu et al., 2021), while the multifidelity use of a coarse prediction as a reference is motivated by Lu et al. (2022). We use a Cartesian product DeepONet whose branch receives the parameters concatenated with a stored LF training field and whose trunk receives target grid coordinates. A nearest neighbour search in standardized parameter space selects that field. The network learns a residual relative to its interpolated version, which is then added back to form the fine prediction. Retrieval is the deployment adaptation: a new query requires parameters only and no new coarse solve. The rationale is to exploit a nearby physical field without first learning a separate coarse surrogate. If an exact LF match exists at a training input, retrieval supplies that match, whereas an unseen query uses a neighboring input. This difference in reference quality remains a limitation. The implemented residual model is also distinct from the composite multifidelity architecture of Howard et al. (2023).

M7: POD GP. Basis emulation represents a large field through a small set of coefficients (Higdon et al., 2008). M7 retains this principle, constructing a POD basis by singular value decomposition of centered HF training fields. A Gaussian process maps parameters to each retained coefficient, and the predicted coefficients are projected back through the basis and added to the training mean field. Our implementation uses an RBF kernel with automatic relevance determination and shares covariance hyperparameters across coefficient outputs. It does not implement the full Bayesian hierarchy of the cited model. M7 is deliberately fine only: no coarse field enters basis construction, fitting, or inference. It supplies a compact alternative when the field varies mainly in a few directions or when learning a coarse correction is less effective than learning the fine coefficients directly.

M8: Slice attention corrector. Transolver motivates grouping spatial cells into learned slice tokens (Wu et al., 2024). We retain pooled spatial summaries but change the attention operation. Original Transolver normalizes each cell’s assignments across slices, applies attention between slice tokens, and maps back using the assignments. Our pooling weights instead normalize across cells, and cell queries attend directly to the pooled slice tokens. The model receives the current field, coordinates with sinusoidal features, and FiLM conditioning on the parameters (Perez et al., 2018). A coarse surrogate ensemble supplies its initial state, with out of fold predictions for training cases. A head initialized to zero predicts additive corrections through six damped updates with shared weights. Training penalizes spatial squared error and Fourier magnitude discrepancy at each update, plus a nonzero update when started at the target field. This encourages an equilibrium without guaranteeing contraction. The adaptation tests whether pooled global interactions can refine a predicted coarse field using parameters alone at deployment. Its attention and iterative recipe differ from the published Transolver.

M9: ConvNeXt corrector. M9 retains ConvNeXt’s spatial and channel mixing blocks (Liu et al., 2022) and adapts them to the same correction task as M8. It uses a custom U shaped encoder and decoder with skip connections (Ronneberger et al., 2015), three downsampling stages and a bottleneck, and FiLM parameter conditioning (Perez et al., 2018). Unlike M3, which generates a

field from coordinates and parameters after transfer training, M9 receives the current coarse estimate as a spatial input and outputs an additive update. Its head is initialized to zero, making the initial refinement the identity. The coarse ensemble construction, six damped updates, shared weights, and spatial, spectral, and target state penalties follow the M8 recipe. The rationale is to offer local processing at several resolutions as an alternative to pooled attention when correcting the same kind of coarse prediction. This is a ConvNeXt based iterative corrector, not the original classification model.

The additional baseline adaptations below use the same distinction between a source idea and a complete published method. In particular, B1, B5, and B8 omit central probabilistic components of co kriging, DMFAL, and IFC. Their scores describe the implemented references, not faithful reproductions of those complete methods. Common evaluation cases make the scores comparable while leaving architecture and resource effects coupled.

B1, B2, and B11: classical multifidelity field references. B1 applies an autoregressive discrepancy idea (Kennedy & O’Hagan, 2000) to POD GP field models. It fits one scalar coupling coefficient in field space, constructs a separate POD GP for residuals, and combines posterior means at deployment. This is a plug in approximation rather than joint probabilistic co kriging. B2 introduces nonlinear dependence on LF coefficients (Perdikaris et al., 2017). Its HF coefficient kernel combines input dependent coupling, an LF coefficient kernel, and a discrepancy kernel. It shares covariance hyperparameters across HF outputs and propagates marginal LF uncertainty through Monte Carlo samples. B11 is a cheaper affine alternative inspired by multifidelity surrogate work (Zhang et al., 2018). One slope and one intercept are fitted jointly over HF training cases and pixels, using observed LF fields when paired and emulated LF fields otherwise. They are then applied to the LF POD GP prediction. It does not fit an affine mapping separately for each mode. Paired training rows can use observed LF fields, whereas deployment uses predicted LF fields.

B3, B4, and B5: neural fusion references. B3 uses the composite correlation idea of Meng & Karniadakis (2020). A tanh MLP maps parameters and coordinates to an LF value. This stage is frozen before linear and nonlinear HF networks are trained on the original inputs and LF prediction, with weight decay on the nonlinear branch. B4 follows the composite DeepONet idea (Howard et al., 2023), predicting fixed coarse sensors with an LF DeepONet and feeding parameters plus sensors to linear and nonlinear HF DeepONets. The LF stage is frozen and the HF branches are trained jointly in a subsequent stage. B5 takes hierarchical latent transfer as inspiration from DMFAL (Li et al., 2022a). It uses two encoders and linear field decoders, with the second encoder receiving parameters and the first latent vector. Joint LF and HF squared losses replace the probabilistic objective. Both the variational last layer and active acquisition are omitted. The name latent MF network reflects these substantial departures.

B6 and B7: alternative decoder and probabilistic models. B6 uses NOMAD’s nonlinear decoding principle (Seidman et al., 2022). A parameter MLP produces a latent vector and a coordinate conditioned MLP decodes the field. The model is implemented independently, with multifidelity learning supplied by LF pretraining and HF finetuning. B7 directly imports MFRNP’s upstream MultiFidelityModel (Niu et al., 2024). Its adapter changes data loading, normalization, resizing, and training. A scalar mean and standard deviation from the first fidelity normalize inputs, while outputs are scaled per fidelity. The longest grid side is capped at 256, rectangular grids are supported, and bilinear antialiased resizing replaces the upstream transform. Batch size is limited by the smallest fidelity, with independent shuffling across fidelities. When HF training has at most ten examples, all rows are retained and the context fraction range changes from $[0.20, 0.25]$ to $[0.40, 0.60]$. Hidden width is 128, latent width is 64, and there are three hidden layers. Adam uses learning rate and epsilon of 10^{-3} . HF likelihood receives weight two and LF likelihood weight one. Latent summaries are saved from the final training batch. These are protocol adaptations of an upstream model, rather than a reproduction of its published experiment.

B8, B9, and B10: fidelity and transfer FNO references. B8 forms latent spatial channels with an FNO and mixes them with weights predicted from $[\ell, \ell^2]$, the fidelity label and its square. These two features condition the spatial basis weights on fidelity. It uses maximum absolute scaling per fidelity, the nearest registered scaler for an intermediate fidelity, LF warm up, and joint LF and HF finetuning. IFC motivates fidelity dependent representation (Li et al., 2022b), but its neural ODE

and Gaussian process are absent. The descriptive name fidelity basis FNO makes this distinction explicit. B9 and B10 are two archived entries for parameter concatenation FNO transfer (Lyu et al., 2023). In the inspected release, their model files are byte identical and their training scripts differ only in repository paths and reported identifiers. The labels FNO transfer I and II distinguish saved result entries, not independent baseline families. Their different saved scores remain unexplained by the inspected source. Both are retained to make the archived comparison auditable, but neither their number nor their score difference is evidence for architectural diversity or a reproducible recipe difference. For ERA5 and the four additional reaction diffusion datasets, the identical recipe is fitted once and its predictions are explicitly reused for B10, with no additional training cost.

B.2 TRAINING SETTINGS AND DATA USE

Table 4 summarizes the library schedules. For M1, M2, M3, and M5, AdamW uses batch size at most 16, weight decay 10^{-5} , gradient clipping at 1, and cosine decay to 10^{-6} . Initial learning rates are 10^{-3} for pretraining and 3×10^{-4} for HF finetuning. The FNO recipe uses width 64, four blocks, and up to 12 Fourier modes per direction, subject to the grid cap. M4 also uses AdamW but starts both its LF members and residual stage at 10^{-3} . M6 uses width 256 and five affine layers of that width in each branch and trunk specification, including the latent output, with ReLU and Adam at 10^{-3} . Detailed grid dependent definitions and fitted hyperparameters are retained with the model implementations and metadata.

Table 4: **Library training schedules.** Epochs count passes through the relevant training table. Optimizer steps and repeated refinement steps are different quantities. Each final surrogate uses a single seed, with internal coarse ensemble initializations prescribed by its recipe.

Models	Nominal training procedure
M1, M3, M5	2,500 LF epochs, followed by 2,500 HF epochs.
M2	2,500 epochs over pooled fidelity pairs, followed by 1,250 HF epochs.
M4	Five LF FNOs and one HF residual FNO, each trained for 2,500 epochs.
M6	125,000 Adam iterations on HF residual targets.
M7	HF basis construction and coefficient GP fitting, without an epoch schedule.
M8, M9	Coarse members request 30,000 optimizer steps, rounded to whole epochs. Each corrector uses 6,000 optimizer steps and six refinement updates per prediction.

M1, M3, M4, M5, and M6 use the lowest registered LF level. M2 pools pairs across registered fidelities, M8 and M9 predict the finest registered LF level, and M7 uses HF fields only. ERA5 consequently uses level one for the transfer and retrieval recipes and level eight for the iterative correctors. For M8 and M9, five LF folds each contain four members: plain and heteroscedastic variants with two initializations each. Training cases receive the mean from their held out fold. All query cases use the four members of fixed fold zero. Corrector checkpoints are selected using training only validation. M7’s basis uses only HF base training fields. Its original estimator state was not saved, so the common prediction export reconstructs that training only fit and restores target scaling.

Baselines use the settings of the evaluated benchmark implementations, with the adaptations above. Training, validation allocations, and computation vary across recipes. For example, on Darcy, B3 to B5 use 360 fine rows for fitting and 40 for internal validation, while B1, B2, and the transfer recipe fit 400 rows. Both allocations consume 400 fine labels before ensemble fitting. The total fine data budget is $N_H^{\text{base}} + K$, plus any additional HF labels used for tuning, where N_H^{base} counts base training labels and K counts ensemble fitting examples. Final evaluation labels are separate. Computation includes every candidate and coarse member trained or evaluated, even if its final weight is zero. Some timing records measure prediction export rather than training, preventing a common training cost estimate. Thus common evaluation cases do not establish equal resources or isolate a multifidelity advantage over a matched fine only ensemble.

C ENSEMBLE IMPLEMENTATION

C.1 NUMERICAL FITTING

The Selected model, Inverse error mixture, and Fitted mixture use the same K input and fine field pairs. Their definitions appear in Section 3. For joint fitting, the empirical error matrix is symmetrized and divided by $\max(\text{tr}(\hat{C})/M, 10^{-30})$. Here the trace sums diagonal errors and $M = 9$ counts models. This positive scaling changes the objective magnitude without changing the matrix condition number or the exact minimizer. SLSQP uses the analytic gradient, bounds $0 \leq w_m \leq 1$, the unit sum constraint, tolerance 10^{-12} , and at most 300 iterations. Uniform weights and the lowest diagonal singleton initialize separate solves. Finite solutions with unit sum residual below 10^{-6} are clipped to nonnegative values and renormalized. Feasible starting points remain candidates, and the lowest fitting objective determines the retained weights. These checks establish numerical feasibility, not generalization to new inputs.

Uniform averaging, used as a control in Table 7, assigns every member weight $1/M$. Forming the error matrices costs $O(KM^2P)$ for P values per field. Streaming predictions and keeping only these matrices requires $O(KM^2)$ storage. These costs exclude surrogate training and inference. The companion package preserves additional regularization, pair fitting, and coarse input refresh experiments separately from the three main rules.

C.2 FROM A TRAINED SURROGATE LIBRARY TO A DEPLOYED ENSEMBLE

Algorithm 1: train models, fit the ensemble, and predict

Input: LF/HF training tables, K reserved fine cases, fixed model recipes, and one prescribed rule: Selected model, Inverse error mixture, or Fitted mixture.

1. Check input identities and working grids within the dataset. Train the surrogate models using training data only. Record unresolved units or simulation metadata.
2. Predict fields for the fitting inputs and compute $G_1, \dots, G_K$.
3. Fit the prescribed rule on all K cases using the definitions in Section 3.
4. Save preprocessing, model identities, and fitted weights.

Predict: combine model predictions at new parameters using Equation (1).

Evaluate: score reserved fine answers only after saving these choices.

The algorithm separates surrogate training from ensemble fitting. The bundled program implements steps 2 to 4 and combines saved model predictions for new inputs. Dataset adapters and base model training remain separate components. The three rules use the same fitting and evaluation cases within each comparison.

D COMPLETE RESULTS AND FITTING BUDGETS

D.1 INDIVIDUAL MODELS, BASELINES, AND ENSEMBLES

Table 1 reports all nine library members and the three ensemble rules on every dataset. Table 5 completes that comparison with individual additional baseline errors. The latter include the parameter kNN and training mean controls, which are outside the ranked B1 to B11 collection. Every ranked method covers all 22 datasets. Errors average the same evaluation cases within each partition, followed by the three partitions. ERA5 uses its separate seven case evaluation set. Full precision values, per case errors, and replay metadata are retained in the companion package.

Table 5: **Additional baseline errors across all 22 datasets.** Mean relative L^2 errors in percent. B1: Autoregressive POD GP. B2: Nonlinear POD GP. B3: Composite MF MLP. B4: Composite MF DeepONet. B5: Latent MF network. B6: Nonlinear decoder transfer. B7: MFRNP adapter. B8: Fidelity basis FNO. B9 and B10: FNO transfer I and II. B11: Affine POD GP. B12: Parameter kNN. B13: Training mean. Mixtures use five fitting examples and the same evaluation cases. A dash marks an unavailable additional control, not a missing ranked method. B9 and B10’s implementation relationship is described in Appendix B.1. Imported data are from MFRNP (Niu et al., 2024) and ERA5 reanalysis (Hersbach et al., 2020). Daggers identify the data quality exclusions in Appendix E.1.

Dataset	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	Inverse error mixture	Fitted mixture
Darcy	3.37	3.66	4.03	6.66	5.91	5.2	12.4	5.55	5.27	5.28	3.6	21.5	38	2.55	2.33
Eikonal	3.49	2.99	3.12	3.35	3.38	2.65	3.62	2.21	2.33	2.34	4.88	5.51	31.8	1.76	1.76
Helmholtz	12.3	15.2	11.5	7.84	8.12	99.6	58.6	99.7	4.12	4.12	15.9	25.7	92	6.89	6.55
Poisson I	0.00731	0.00531	0.181	0.161	0.1	0.479	6.79	0.239	0.0804	0.0806	2.68	7.51	31.9	0.008	0.00788
Poisson II [†]	0.0128	0.0306	0.635	1.72	0.849	3.93	11.7	82.1	0.355	0.357	2.01	12.3	38.5	0.0125	0.0129
Pressure Poisson [†]	0.457	0.472	0.07	1.53	0.127	0.446	1.05	0.684	0.0639	0.0645	0.681	0.196	1.58	0.0769	0.0802
Heat I	0.326	0.327	0.0696	0.125	0.069	0.432	1.29	0.275	0.0207	0.0206	1.09	1.35	12.4	0.00593	0.0062
Heat II	0.312	0.312	0.111	0.183	0.0721	0.219	7.4	0.32	0.0202	0.0201	0.405	0.856	13.2	0.00437	0.00435
Cavity	0.307	0.409	23.9	4.99	0.0419	24.1	55.7	0.129	0.622	0.666	37.2	0.077	17.1	0.142	0.142
Navier Stokes	2.8	2.48	2.08	4.01	3.71	7.08	25.7	1.62	1.52	1.52	21.6	3.27	41.2	1.52	1.5
Rayleigh Bénard	0.283	0.295	0.0542	0.0588	0.0134	0.0613	4.5	0.78	0.0113	0.011	0.715	0.0575	9.1	0.00286	0.00269
Burgers	2.16	2.5	1.54	8.14	1.22	15.7	27.7	0.928	0.562	0.562	11.7	0.677	73.2	0.513	0.502
Euler	5.6	5.54	6.56	6.16	6.15	8.81	9.15	7	6.52	6.43	6.25	6.21	32.2	5.64	5.73
Porous medium	1.15	0.859	0.299	1.31	0.772	0.918	33.7	5.14	0.0718	0.0718	1.98	1.92	31.9	0.0262	0.0263
Allen Cahn	55.5	98.6	59.9	44.8	41.2	94.4	69	48.9	55.8	55.8	55.4	78.3	100	37.9	34.6
Cahn Hilliard I [†]	29.7	27.8	53.5	48.7	27.6	61.1	74.7	14.2	13.4	13.3	72.3	20.6	99.9	11.3	10.7
Cahn Hilliard II	64.8	83.5	85	53.7	49.8	72.4	66.7	54.4	54	54	64.8	78	100	47.2	49.3
Fisher KPP	6.86	6.86	6.48	4.01	3.45	7.44	4.08	4.74	3.24	3.24	6.86	5.67	6.86	2.53	2.3
Phase field crystal	83.5	86.4	90.7	87.5	87.8	152	89.4	102	86.8	86.8	83.5	86.7	87.5	83.8	83.3
Shallow water	1.14	1.02	0.467	0.878	0.691	0.883	5.59	0.806	0.403	0.403	2.94	1.22	25.1	0.357	0.389
Wave	27.7	24.9	3.3	42.3	46.6	5.39	93.9	25.9	6.6	6.6	50.3	42.8	98.7	4.39	4.27
ERA5	5	6.74	45	16.8	5.05	23.9	19.5	5.15	11.9	11.9	8.45	5.16	6.74	5.81	6.49

D.2 LIBRARY QUALITY AND THE BENEFIT OF AVERAGING

Two comparisons separate the quality of the available library from the benefit of combining it. The best library reference minimizes partition averaged error over M1 to M9 separately for each dataset. The best additional baseline minimizes the corresponding error over B1 to B11. Both use evaluation answers retrospectively and are stronger references than the deployable Selected model, which uses only fitting examples. Table 6 reports these comparisons on the 21 PDE datasets. Table 2 additionally includes ERA5 in its dataset ratios and Elo ranking.

A still stronger control chooses a potentially different best model in each partition using its evaluation answers. Against this reference, the fitted mixture at $K = 5$ has class ratio 0.933 and dataset ratio 1.006, with 13 improvements and 6 losses greater than 1%. Thus its advantage over retrospective model choice depends on both the reference and aggregation.

Table 6: **Library quality and mixture gains on the 21 PDE datasets.** Ratios are geometric means. Wins and losses require a relative difference greater than 1%. Best library and best additional baseline refer to the retrospective per dataset minima described above.

Comparison	Class ratio	Dataset ratio	Wins	Losses
Best library / best additional baseline	0.766	0.834	15	5
Fitted mixture / best library	0.923	0.989	13	6
Inverse error mixture / best library	0.927	1.002	12	7
Fitted mixture / best additional baseline	0.706	0.825	13	7
Fitted mixture / Selected model	0.884	0.923	15	5

D.3 FITTING BUDGETS AND UNIFORM AVERAGING

Table 7 compares the main rules with uniform averaging on the 21 PDE datasets. Table 8 varies the fitting budget while keeping the models and case partition protocol fixed. Each ratio compares methods that receive the same number of fitting examples. These summaries measure sensitivity to ensemble fitting, not independent training seed variation.

Table 7: **Weighting the same nine model library.** Ratios compare with the Selected model at $K = 5$ on 21 PDE datasets. Lower is better. Wins and losses require changes larger than 1%.

Method	Class balanced	Equal dataset	Wins	Losses
Selected model	1.000	1.000	0	0
All pairs FNO	1.333	1.502	3	15
Uniform (all nine)	5.919	4.944	2	19
Inverse error mixture	0.888	0.935	13	5
Fitted mixture	0.884	0.923	15	5

Table 8: **Increasing the fitting budget on the 21 PDE datasets.** Class aggregated error ratios. The models remain fixed, and every comparison uses the same allowance of K additional fine fields.

K	Inverse error mixture / Selected model	Fitted mixture / Selected model	Fitted mixture / Inverse error mixture
5	0.8876	0.8840	0.9958
10	0.8921	0.8765	0.9826
20	0.8900	0.8692	0.9766

D.4 ELO RANKING

Each pair of methods plays one comparison per dataset using its unrounded partition averaged error. Lower error wins and an absolute difference below 10^{-12} is a tie. The nine library members, eleven baseline entries, and three rules all enter on the same 22 datasets. Ratings start at 1500. For ratings R_a, R_b , the expected score and update are

$$p_a = \frac{1}{1 + 10^{(R_b - R_a)/400}}, \quad R'_a = R_a + 32(s_a - p_a), \quad R'_b = R_b - 32(s_a - p_a). \quad (7)$$

Here s_a is one for a win, zero for a loss, and one half for a tie. Primes denote updated ratings. We reset ratings and independently shuffle comparisons 500 times, then report the mean final rating. The code fixes the random orders. Sorting uses unrounded ratings and display uses integers. Datasets receive equal weight. Elo summarizes how often a method wins rather than the magnitude of improvement, statistical significance, or training cost.

E ROBUSTNESS CHECKS AND ERA5 DETAILS

E.1 SENSITIVITY TO COMPARISON SCOPE

The following diagnostics reuse the original 17 PDE tasks within the expanded collection. They retain the trained models, predictions, and fitting cases. The 14 task data quality subset excludes Poisson II, Cahn Hilliard I, and Pressure Poisson for the properties described in Appendix A. A second subset removes Heat II and Burgers, the two tasks whose B8 case correspondence was established through checkpoint evaluation. Their intersection contains 12 tasks. Class aggregation normalizes over represented classes. These retrospective sensitivity checks do not certify solver validity or provide independent replication.

Table 9: **Sensitivity to dataset inclusion.** Each cell gives the PDE class ratio followed by the dataset ratio. N counts tasks. Library and baseline refer to retrospective best models after averaging partitions. Both mixtures use five fitting examples.

Scope	N	Library / baseline	Fitted mixture / Selected model	Inverse error mixture / Selected model
Diagnostic set	17	0.764 / 0.820	0.876 / 0.918	0.882 / 0.924
Data quality subset	14	0.747 / 0.810	0.873 / 0.900	0.875 / 0.908
Without Heat II and Burgers	15	0.790 / 0.865	0.872 / 0.935	0.872 / 0.940
Intersection	12	0.776 / 0.865	0.869 / 0.918	0.865 / 0.926

Gains over the Selected model persist across these scopes. Superiority to the best individual library member is less stable. On the 15 task subset, the fitted mixture has class ratio 0.916 and dataset ratio 1.020 against that reference, improving under one aggregation and worsening under the other. The complete subset and budget tables remain in the companion package.

Source hashes also identify identical M8 and M9 prediction archives on Darcy. These entries do not supply distinct prediction diversity on that dataset. Their weights can be redistributed without changing a fitted combined field, while inverse error weighting is affected by duplication because each copy receives weight before normalization. Effective model count therefore measures weight concentration over entries, not independent mechanisms. We retain the evaluated library and disclose this property rather than interpreting those weights as unique measures of model usefulness.

E.2 ERA5 PROTOCOL AND CONTROLS

All nine library models use ERA5 training inputs disjoint from the ten fitting pool and seven evaluation inputs (Hersbach et al., 2020, Niu et al., 2024). M8 and M9 use 49 fine training cases for fitting and six for internal validation, with four coarse predictions per case assembled by the procedure in Appendix B.2. The twelve completed baseline entries represent eleven independent fits because B10 reuses B9, and the training mean provides a separate control. Ranked baselines B1 to B11 all have scores on the same seven evaluation cases.

Table 10: **ERA5 mixtures and controls at two fitting budgets.** Mean relative L^2 errors in percent. At $K = 5$, three fitting selections use subsets of the same ten case pool. At $K = 10$, they repeat the complete pool. Both budgets reuse the same seven evaluation cases and trained models.

Method	$K = 5$	$K = 10$
Inverse error mixture	5.8146	5.8358
Fitted mixture	6.4877	6.4736
Selected model (POD GP)	6.7369	6.7369
Training mean field	6.7369	6.7369
Uniform mixture	6.3730	6.3730
All pairs FNO	17.6992	17.6992
Fine only FNO	23.3336	23.3336

POD GP is the best library member, but its output is constant across all 17 reserved inputs and matches the training mean to displayed precision. Every neural library member has higher error. The inverse error mixture nevertheless improves on that reference, while autoregressive POD GP and fidelity basis FNO remain more accurate than either mixture (Table 5). Figure 4 shows that the inverse error mixture improves six evaluation cases and the fitted mixture improves five. These seven shared cases do not support a significance claim or independent training replicates.

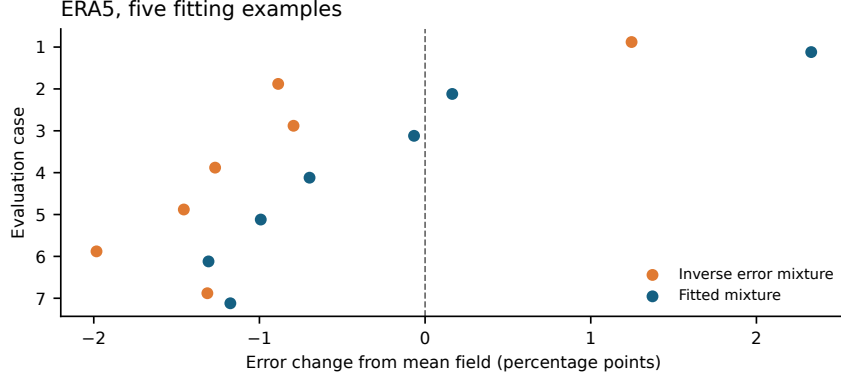


Figure 4: **ERA5 gains by evaluation case.** Change from the training mean in relative L^2 percentage points, averaged over three $K = 5$ fitting selections. Negative values favor the mixture. Case numbers identify archive rows rather than dates.

Predictions and losses use a 128×256 working grid, while the native fine array has 721×1440 values. The input variables describe supplied simulation drivers, not an established forecast origin. Absolute units, calendar coordinates, and the target transformation remain unverified from the source arrays. Relative temperature error changes under an additive unit offset, and the unweighted grid metric is not an area weighted global metric. These results concern the registered field emulation task, not global climate skill or operational forecasting.

E.3 ADDITIONAL EVALUATION INPUTS FOR HEAT I

This supplementary check uses the existing Heat I library on 40 new fitting pool cases and 128 new evaluation cases. Three fitting selections share those evaluation inputs. Table 11 reports the three main rules separately from the 22 dataset comparison. This single PDE check predates the direct loss control in Appendix F.1 and does not establish generalization to new problem families.

Table 11: **Additional Heat I inputs.** Mean relative L^2 errors in percent. Three fitting selections share the same 128 evaluation inputs and trained models.

K	Selected model	Inverse error mixture	Fitted mixture	All pairs FNO
5	0.008638	0.005300	0.005454	0.011171
10	0.008638	0.005286	0.005359	0.011171
20	0.008638	0.005270	0.005296	0.011171

F MATHEMATICAL EXPLANATION AND RELATED SYSTEMS

F.1 ERROR GEOMETRY AND THE FITTING OBJECTIVE

For a fixed library, Equation (6) expresses expected squared relative error as $R(w) = w^\top C w$, with $C = \mathbb{E}[G(x, y)]$. The error product matrix is uncentered, so shared bias remains part of the risk. Its diagonal describes individual accuracy and its cross terms describe error alignment (Breiman, 1996, Krogh & Vedelsby, 1994). Writing $d_m = \hat{C}_{mm} > 0$, minimization of the diagonal approximation $\sum_m d_m w_m^2$ under simplex constraints gives $2d_m w_m = \lambda$. The scalar λ enforces the unit sum constraint, so normalization yields $w_m \propto d_m^{-1}$. This recovers the inverse error mixture before its numerical floor, without assuming independent model errors. If some $d_m = 0$, an optimum is supported on the zero error entries.

For one case, let e_m be model m 's normalized error field and $r_m = \|e_m\|_Q$ its magnitude. The improvement relative to errors of the same sizes that all point in one direction is

$$\left(\sum_m w_m r_m \right)^2 - w^\top G w = 2 \sum_{m < n} w_m w_n (r_m r_n - \langle e_m, e_n \rangle_Q) \geq 0. \quad (8)$$

Here G is the same case's error product matrix, and $m < n$ counts each model pair once. Expansion gives the equality and Cauchy Schwarz gives nonnegativity. The identity explains how errors in different locations or directions can help either mixture. Negative correlation is not required, and the identity does not establish that a small fitting set will discover useful weights.

The fitting criterion can change the preferred mixture (Acar & Rais-Rohani, 2009, Acar, 2015). Our joint fit minimizes mean squared relative error, whereas reporting uses

$$L(w) = \mathbb{E} \left[\sqrt{w^\top G(x, y) w} \right]. \quad (9)$$

Here L averages the relative L^2 error over inputs and targets, taking the square root separately for each field. Jensen's inequality gives $L(w) \leq \sqrt{R(w)}$ for fixed weights, but does not equate their minimizers. For example, two equally likely scalar cases with normalized model errors $e_A = (0, 2)$ and $e_B = (1.1, 1.1)$ give

$$L(t) = 1.1 - 0.1t, \quad R(t) = 1.21 - 0.22t + 1.01t^2, \quad (10)$$

where $t \in [0, 1]$ weights model A and $1 - t$ weights B. The minimum of L is at $t = 1$, while R is minimized at $t = 11/101$, where $L = 110/101$. Thus fitting squared error can be about 8.9% worse on the reported metric even with unlimited fitting cases. Constant spatial fields give the same example.

Direct fitting of the reporting metric. A standard convex alternative fits the unsquared error directly:

$$\min_{w \in \Delta_M} \frac{1}{K} \sum_{i=1}^K \sqrt{w^\top G_i w} = \min_{w \in \Delta_M} \frac{1}{K} \sum_{i=1}^K \|A_i w\|_2, \quad A_i^\top A_i = G_i. \quad (11)$$

Here A_i factors the error matrix of fitting field i , K counts fields, and Δ_M contains nonnegative weights summing to one. Each term is a norm of a linear map. Introducing upper bounds t_i with $\|A_i w\|_2 \leq t_i$ gives a second order cone problem. We evaluate this loss control on 51 partitions of the original 17 PDE tasks at $K = 5$, keeping all predictions and case assignments fixed. The weights were saved before their new scores were computed, but the predictions had previously been inspected, making this a retrospective diagnostic separate from the three main rules.

CVXPY 1.9.2 and Clarabel 0.11.1 solve the scaled problem with absolute and relative gap tolerances 10^{-10} and feasibility tolerance 10^{-9} . All 51 solves returned optimal status. Eigenvalues below zero are clipped only within 10^{-12} of spectral scale, with larger violations rejected. The worst relative eigenvalue was approximately -1.04×10^{-18} and the largest simplex residual before clipping and renormalization was 4.15×10^{-11} . Source identities, reference error replay, and solver records are retained with the locked fits.

Table 12: **Fitting the reporting metric directly.** Ratios compare with the Selected model on the 17 PDE diagnostic tasks at five fitting examples. Wins and losses require more than 1% change after averaging three partitions. The direct loss fit is a supplementary control, not a fourth main rule.

Method	Class ratio	Dataset ratio	Wins	Losses
Inverse error mixture	0.882	0.924	11	4
Fitted mixture	0.876	0.918	12	5
Relative L^2 fit	0.859	0.898	12	5

Direct relative L^2 fitting reduces class aggregated error by 2.0% and dataset aggregated error by 2.1% relative to the original squared error fit. Nine datasets improve by more than 1%, none worsens by that margin, and eight remain within 1%. Against the Selected model, 12 improve and five worsen. The result identifies a consequential loss choice without establishing a new weighting principle or independent validation.

F.2 WHAT ESTIMATING THE ERROR MATRIX GUARANTEES

Proposition 1 (Error estimation and mixture risk). *If $\max_{m,n} |\hat{C}_{mn} - C_{mn}| \leq \epsilon$ and a feasible $\hat{w}$ has empirical objective within $\delta \geq 0$ of the simplex minimum, then*

$$R(\hat{w}) - R(w^*) \leq 2\epsilon + \delta, \quad (12)$$

where w^* minimizes population squared relative risk over the simplex.

Here ϵ bounds matrix estimation error and δ bounds numerical optimization error. For every feasible w , nonnegative unit sum weights give $|w^\top (\hat{C} - C)w| \leq \epsilon \sum_{m,n} w_m w_n = \epsilon$. Applying this once at $\hat{w}$ and once at w^* , with empirical near optimality between them, proves the result. The bound separates estimation from optimization error but does not certify small ϵ from five fields. Standard concentration arguments (Hoeffding, 1963) require bounded error products and independent fitting fields, assumptions not established here. Spatial pixels are not independent fitting examples. The companion package retains the longer concentration calculation, whose bound is weaker than a trivial bounded risk guarantee at the reported budgets. Neither calculation establishes near optimality for the unsquared metric L .

F.3 RELATIONSHIP TO THE CLOSEST PRIOR SYSTEMS

Model weighting, automated surrogate selection, and multifidelity field emulation are established ideas. Table 13 identifies these precedents and the scope of our integration. The contribution is the adapted portfolio, common parameter input interface, and experiments with counted fitting examples. The table compares formulations rather than measured performance. Direct comparisons with complete engineering automation systems would require matched training, tuning, and information budgets.

Table 13: **Prior systems and the scope of this study.** These workflows were not all evaluated on the present tasks. Adapted baseline implementations do not substitute for complete system comparisons.

Predecessor	Established contribution	Present investigation
Engineering ensembles (Acar & Rais-Rohani, 2009, Viana et al., 2009)	Validation based weighting and model selection.	Neural field library with several fidelity strategies and counted fitting examples.
Functional output benchmark (Brunel et al., 2025)	Unified comparison of multifidelity field emulators.	Heterogeneous neural and reduced basis candidates combined from saved fields.
Stacking and AutoGluon (Wolpert, 1992, Erickson et al., 2020)	Automated library combination.	Whole field loss over a fixed portfolio of fidelity strategies.
EL MFS and AQBMF (Wang et al., 2024, Lee et al., 2026)	Adaptive LF fusion and MF strategy selection.	Diverse field architectures with parameter only inference.
MAESTRO (Kim et al., 2026)	LF and discrepancy families with screening and leave one out selection.	Direct, correction, and fine only field models.
LANE SI (Borisova & Nikitin, 2024)	Neural and climatological spatial forecast ensembles.	Parametric emulation with supplied multifidelity simulations.
HyperNOs (Ghiotto, 2026)	Automated operator training and hyperparameter search.	Fitting weights across different fidelity use strategies.
MFRNP and MF DeepONet (Niu et al., 2024, Howard et al., 2023)	Multifidelity field architectures.	Common ensemble fitting over multiple adapted families.

G REPRODUCIBILITY, AI USE, AND ETHICS STATEMENTS

G.1 REPRODUCIBILITY STATEMENT

We will release the benchmark datasets and the code used to train and evaluate all baseline implementations, the nine surrogate models, and the ensemble methods, together with data preparation scripts, fixed splits, experiment configurations, and scripts for reproducing the reported tables and figures.

We retain the datasets, training and simulation code, data preparation routines, and experiment records for all 22 datasets in the benchmark and experiment archives. The manuscript package supplies numerical snapshots, plotting and table scripts, a workbook, and a portable program for fitting ensembles from saved predictions and combining new predictions without their evaluation answers. Input identities, source hashes, saved weights, and replay checks connect these artifacts to the reported scores. The verified clean rebuild covers ensemble fitting, saved prediction evaluation, tables, and figures. It does not retrain the library or rerun the simulation solvers. The companion supplement indexes enlarged dataset panels, exploratory experiments, and detailed audit records omitted from the condensed appendix.

G.2 AI USE STATEMENT

Generative AI assisted hypothesis development, experiment design, coding, provenance inspection, result interpretation, mathematical analysis, literature search, writing, visualization, and references. Reported values come from experiment artifacts and are checked automatically. Scientific claims and source attributions remain the authors' responsibility.

G.3 ETHICS STATEMENT

Inaccurate field predictions can mislead engineering or climate decisions. Benchmark error alone establishes neither physical validity nor operational suitability. Dataset limitations and the scope of the ERA5 metric are disclosed in the evaluation protocol and climate appendix. Data provenance and redistribution permissions require review before public release.